
Amazon EventBridge

What is Amazon EventBridge?

Amazon EventBridge is a **serverless event bus** that enables **event-driven architectures** by connecting your applications with AWS services, integrated SaaS partners, and custom apps through real-time events.

Core Components

Component	Description
Event	JSON message containing structured data (e.g., EC2 state change, S3 upload)
Event Source	The origin of the event: AWS services, SaaS apps, or custom apps
Event Bus	Pipeline that receives and routes events
Rule	Filters and matches events based on patterns
Target	Where the event is routed (e.g., Lambda, SQS, SNS)
Schema Registry	Auto-detects and stores structure of events
Archive & Replay	Save and reprocess events for future analysis or debugging

Supported Event Sources

Source Type	Example
AWS Services	EC2, S3, Lambda, CodePipeline, etc.
SaaS Applications	Zendesk, PagerDuty, Segment, etc.
Custom Applications	Your microservices or apps using PutEvents API

Target Destinations

- Lambda
 - Step Functions
 - SNS/SQS
 - ECS Tasks
 - Kinesis Streams
 - API Destinations (external HTTP endpoints)
 - CloudWatch Logs
 - EventBridge Pipes
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Event Filtering (Pattern Matching)

Supports filtering based on fields such as:

- "source" – service generating event
 - "detail-type" – type of event (e.g., "Object Created")
 - "detail" – inner JSON values (e.g., "state": "stopped")
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Example Event Pattern (EC2 instance stopped):

```
{
  "source": ["aws.ec2"],
  "detail-type": ["EC2 Instance State-change Notification"],
  "detail": {
    "state": ["stopped"]
  }
}
```

Use Case: S3 → EventBridge → Lambda

Trigger Lambda when a new object is uploaded to S3:

1. **S3** emits an object-created event.
 2. **EventBridge Rule** matches and routes to Lambda.
 3. **Lambda** processes the event (e.g., logging or triggering a workflow).
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Schema Registry

Feature

Description

Discover Schemas	Automatically infers schemas from incoming events
Download Code Bindings	Generates SDK code in Java, Python, TypeScript
Custom Schemas	You can upload and manage your own schemas

Archive and Replay

Feature	Description
Archive Events	Retain event data for reprocessing
Replay Events	Replay historical events through the same rules
Useful For	Debugging, Replays, Disaster Recovery

Security

- IAM policies for EventBridge actions
 - Resource policies for cross-account access
 - Integration with KMS for encryption at rest
 - DLQs (Dead Letter Queues) supported
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 **Pricing** (as of mid-2025; refer to AWS [Pricing Page](#) for updates)

Feature	Price
PutEvents API	\$1.00 per 1 million events
Events matched and delivered	\$1.00 per 1 million
Schema Registry	Free for AWS Events; paid for custom events
Archive & Replay	Archive: \$0.10 per GB/month Replay: \$0.10 per GB replayed
Event Bus invocations	Billed based on matched events sent to targets
Free Tier	100,000 events per month (PutEvents + matched events)



Example:

If your app emits 2 million custom events/month via PutEvents and 1 million are routed to Lambda:

→ (2M PutEvents + 1M Matched) = **3 million events = \$3/month**

✓ Tips for AWS Solutions Architect Exam

1. Use **EventBridge** for loosely coupled, event-driven architectures.
2. Choose **EventBridge over SNS/SQS** when:
 - You need **event filtering**
 - You want **SaaS integration**
 - You want **multiple rules per bus**

3. **Step Functions** can be used as EventBridge targets for orchestrated workflows.
4. **Schema Registry & Replay** are unique EventBridge features — understand their use.
5. **Custom Events** require the **PutEvents** API + appropriate permissions.

 Comparison Table

Feature	EventBridge	SNS	SQS
Event Filtering	✓ Advanced	✗ Topic-based only	✗
Multiple Consumers	✓ Multiple Rules	✓ Fan-out	✗
DLQ Support	✓ Yes	✗ Manual setup	✓ Native
Event Replay	✓ Built-in	✗	✗
Schema Management	✓ Yes	✗	✗
Delivery Guarantee	At-least-once	Best-effort	At-least-once
